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PART A: VECTOR OPERATIONS

In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from numpy.linalg import eig, det, inv
```

In [21]:
```
df = pd.read_csv("C:\\Users\\tanaa\\Desktop\\student_marks_dataset.csv")
df.head()
```

Out[21]:

	Student_ID	Math	Physics	Chemistry	English	Computer	Total
0	S001	78	85	67	90	88	408
1	S002	65	70	72	68	75	350
2	S003	88	92	85	91	89	445
3	S004	55	60	58	62	57	292
4	S005	90	95	93	89	94	461

In [22]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 7 columns):
 #   Column      Non-Null Count  Dtype
---  ------      --------------  -----
 0   Student_ID  10 non-null     str
 1   Math        10 non-null     int64
 2   Physics     10 non-null     int64
 3   Chemistry   10 non-null     int64
 4   English     10 non-null     int64
 5   Computer    10 non-null     int64
 6   Total       10 non-null     int64
dtypes: int64(6), str(1)
memory usage: 692.0 bytes
```

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In [23]:
```
df.describe()
```

Out[23]:

	Math	Physics	Chemistry	English	Computer	Total
count	10.000000	10.000000	10.00000	10.000000	10.000000	10.000000
mean	73.900000	78.200000	74.10000	76.800000	77.800000	380.800000
std	11.855144	11.735985	10.81614	11.688551	11.726513	55.953949
min	55.000000	60.000000	58.00000	61.000000	57.000000	292.000000
25%	66.000000	70.500000	67.75000	68.500000	73.000000	351.250000
50%	74.500000	77.500000	71.50000	75.000000	77.000000	377.000000
75%	83.250000	87.250000	82.50000	88.500000	87.500000	424.500000
max	90.000000	95.000000	93.00000	91.000000	94.000000	461.000000

In [24]:
```
X = df.drop("Student_ID", axis=1).values
print("Matrix:\n", X)
```

```
Matrix:
 [[ 78  85  67  90  88 408]
 [ 65  70  72  68  75 350]
 [ 88  92  85  91  89 445]
 [ 55  60  58  62  57 292]
 [ 90  95  93  89  94 461]
 [ 72  75  70  74  73 364]
 [ 60  65  63  61  64 313]
 [ 85  88  84  87  86 430]
 [ 77  80  78  76  79 390]
 [ 69  72  71  70  73 355]]
```

In [25]:
```
v1 = X[0]
v2 = X[1]

print("Vector 1:", v1)
print("Vector 2:", v2)
```

```
Vector 1: [ 78  85  67  90  88 408]
```

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In [25]:
```
v1 = X[0]
v2 = X[1]

print("Vector 1:", v1)
print("Vector 2:", v2)
```

```
Vector 1: [ 78  85  67  90  88 408]
Vector 2: [ 65  70  72  68  75 350]
```

In [26]:
```
print("Norm-1:", np.linalg.norm(v1, ord=1))
print("Norm-2:", np.linalg.norm(v1))
```

```
Norm-1: 816.0
Norm-2: 447.33209140413794
```

In [27]:
```
dot = np.dot(v1, v2)
print("Dot Product:", dot)
```

```
Dot Product: 171364
```

In [28]:
```
angle = np.arccos(dot / (np.linalg.norm(v1) * np.linalg.norm(v2)))
print("Angle (radians):", angle)
```

```
Angle (radians): 0.045756570651255206
```

In [29]:
```
cross = np.cross(v1[:3], v2[:3])
print("Cross Product:", cross)
```

```
Cross Product: [ 1430 -1261   -65]
```

In [30]:
```
proj = (np.dot(v1, v2) / np.dot(v2, v2)) * v2
print("Projection of v1 on v2:", proj)
```

```
Projection of v1 on v2: [ 75.74331216  81.56972079  83.90028424  79.23915734  87.39612942
 407.84860395]
```

PART B: MATRIX OPERATIONS

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PART B: MATRIX OPERATIONS

In [31]:
```
XT = X.T
print("Transpose:\n", XT)
```

```
Transpose:
 [[ 78  65  88  55  90  72  60  85  77  69]
 [ 85  70  92  60  95  75  65  88  80  72]
 [ 67  72  85  58  93  70  63  84  78  71]
 [ 90  68  91  62  89  74  61  87  76  70]
 [ 88  75  89  57  94  73  64  86  79  73]
 [408 350 445 292 461 364 313 430 390 355]]
```

In [32]:
```
mul = np.dot(X, XT)
print("Matrix Multiplication:\n", mul)
```

```
Matrix Multiplication:
 [[200106 171364 217961 143008 225696 178277 153252 210576 190944 173823]
 [171364 147058 186893 122642 193648 152877 131484 180599 163814 149122]
 [217961 186893 237660 155945 246175 194397 167147 229637 208263 189573]
 [143008 122642 155945 102346 161532 127557 109680 150683 136654 124394]
 [225696 193648 246175 161532 255052 201367 173172 237879 215764 196400]
 [178277 152877 194397 127557 201367 159010 136723 187836 170355 155067]
 [153252 131484 167147 109680 173172 136723 117580 161513 146496 133350]
 [210576 180599 229637 150683 237879 187836 161513 221890 201243 183183]
 [190944 163814 208263 136654 215764 170355 146496 201243 182530 166148]
 [173823 149122 189573 124394 196400 155067 133350 183183 166148 151240]]
```

In [33]:
```
cov = np.cov(XT)
print("Covariance Matrix:\n", cov)
```

```
Covariance Matrix:
 [[ 140.54444444  138.24444444  116.78888889  130.86666667  133.08888889
   659.53333333]
 [ 138.24444444  137.73333333  112.53333333  132.26666667  134.15555556
   654.93333333]
 [ 116.78888889  112.53333333  116.98888889   94.91111111  107.8
   549.02222222]
 [ 130.86666667  132.26666667   94.91111111  136.62222222  130.06666667
   634.73333333]
```

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Python 3 (ipykernel)

In [34]:
```
print("Determinant:", det(cov))
```

```
Determinant: -3.317154038623629e-08
```

In [35]:
```
print("Determinant:", det(cov))
```

```
Determinant: -3.317154038623629e-08
```

In [36]:
```
try:
    inv_matrix = inv(cov)
    print("Inverse Matrix:\n", inv_matrix)
except:
    print("Matrix not invertible")
```

```
Inverse Matrix:
 [[-1.7592186e+12 -1.7592186e+12 -1.7592186e+12 -1.7592186e+12
  -1.7592186e+12  1.7592186e+12]
 [-1.7592186e+12 -1.7592186e+12 -1.7592186e+12 -1.7592186e+12
  -1.7592186e+12  1.7592186e+12]
 [-1.7592186e+12 -1.7592186e+12 -1.7592186e+12 -1.7592186e+12
  -1.7592186e+12  1.7592186e+12]
 [-1.7592186e+12 -1.7592186e+12 -1.7592186e+12 -1.7592186e+12
  -1.7592186e+12  1.7592186e+12]
 [-1.7592186e+12 -1.7592186e+12 -1.7592186e+12 -1.7592186e+12
  -1.7592186e+12  1.7592186e+12]
 [ 1.7592186e+12  1.7592186e+12  1.7592186e+12  1.7592186e+12
   1.7592186e+12 -1.7592186e+12]]
```

PART C: VISUALIZATION

In [45]:
```
df.set_index("Student_ID").plot(kind="bar", figsize=(10,5))
plt.title("Student Scores Comparison")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()
```

Student Scores Comparison

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Python 3 (ipykernel)

PART D: EIGENVALUES & EIGENVECTORS

In [38]:
```
eigenvalues, eigenvectors = eig(cov)

print("Eigenvalues:\n", eigenvalues)
print("Eigenvectors:\n", eigenvectors)
```

```
Eigenvalues:
 [ 3.75963311e+03  3.28668788e+01  6.11742751e+00  1.25917805e+00
 -8.38210111e-14  3.67855110e-01]
Eigenvectors:
 [[ 1.92230148e-01 -5.85626318e-02 -3.95000392e-01  5.94835601e-01
  -4.08248290e-01  5.32068247e-01]
 [ 1.90910075e-01  7.96104252e-02 -1.48121597e-01  3.03347273e-01
  -4.08248290e-01 -8.22550473e-01]
 [ 1.59892382e-01 -7.92556246e-01 -9.29035294e-03 -4.23632340e-01
  -4.08248290e-01  8.46681216e-03]
 [ 1.82182874e-01  5.73703577e-01 -3.01874689e-01 -6.01858766e-01
  -4.08248290e-01  1.32833118e-01]
 [ 1.87335891e-01  1.80719359e-01  8.54879929e-01  1.10285720e-01
  -4.08248290e-01  1.50321247e-01]
 [ 9.12551371e-01 -1.70855163e-02  5.92898880e-04 -1.70225113e-02
   4.08248290e-01  1.13895123e-03]]
```

PART E: PCA (DIMENSION REDUCTION)

In [39]:
```
# Sort eigenvalues
idx = eigenvalues.argsort()[::-1]
eigenvalues = eigenvalues[idx]
eigenvectors = eigenvectors[:, idx]

# Reduce to 2D
X_reduced = np.dot(X, eigenvectors[:, :2])

print("Reduced Data:\n", X_reduced)
```

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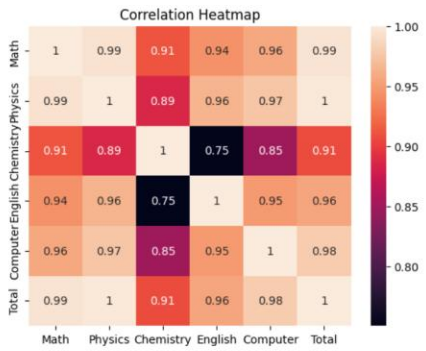
Python 3 (ipykernel)

In [42]:
```
import seaborn as sns

corr = df.drop("Student_ID", axis=1).corr()

sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



In [43]:
```
plt.hist(df["Math"])
plt.title("Math Score Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```

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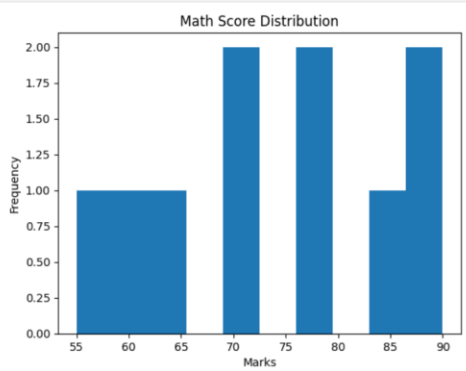
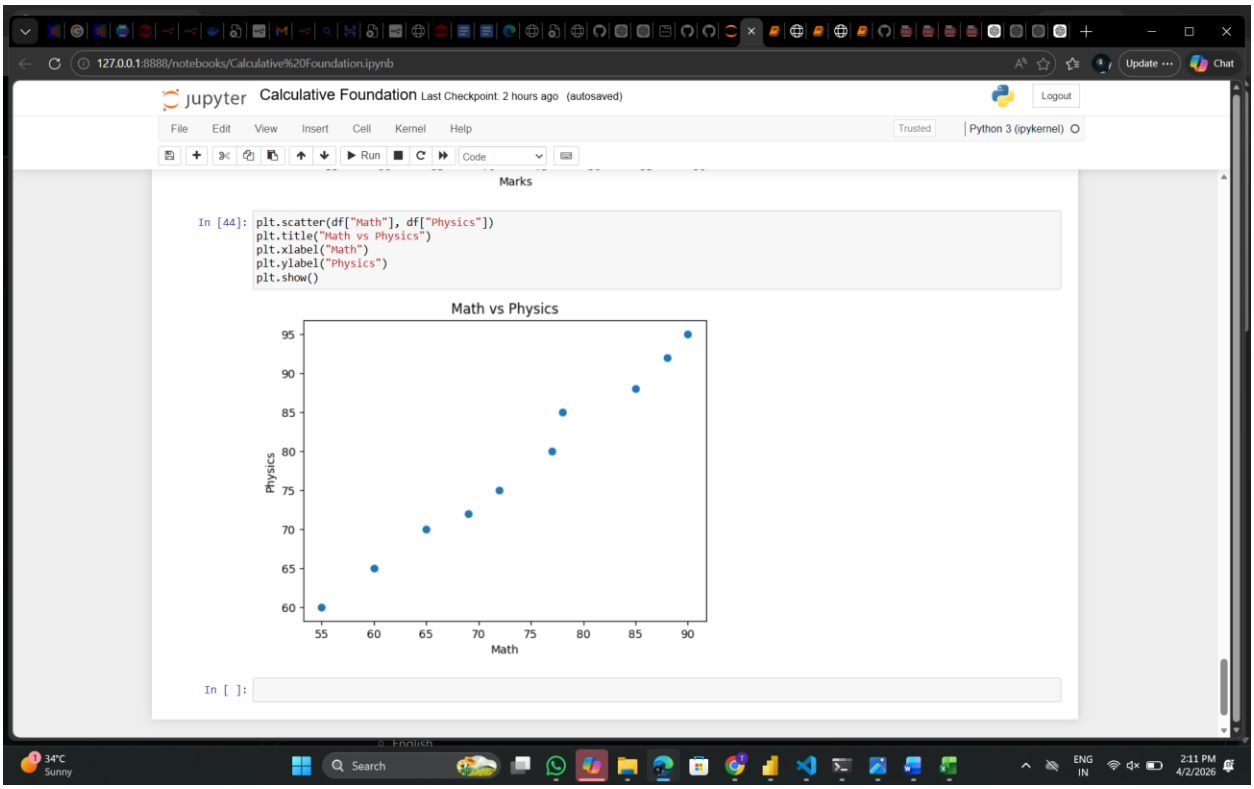
Python 3 (ipykernel)

In [43]:
```
plt.hist(df["Math"])
plt.title("Math Score Distribution")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()
```



In [44]:
```
plt.scatter(df["Math"], df["Physics"])
plt.title("Math vs Physics")
plt.xlabel("Math")
plt.ylabel("Physics")
plt.show()
```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Trusted
Python 3 (ipykernel)
Marks
In [44]:
plt.scatter(df["Math"], df["Physics"])
plt.title("Math vs Physics")
plt.xlabel("Math")
plt.ylabel("Physics")
plt.show()
Math vs Physics
Physics
Math
In [ ]: